



RKV ResiNidhi AI

Rural Cash-Flow Digital Twin & Risk-to-Rescue Engine

An AI-driven platform that predicts 3 – 6month cash flows, detects early financial stress, simulates climate/market shocks, and recommends the minimum intervention needed to prevent risk escalation.

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Problem Statement — Our Interpretation

The challenge is not only “credit scoring”; it is early detection of cash-flow stress before damage happens.

NABARD requirement	What it means in practice	Our design response
3–6 month cash-flow prediction	Banks and field officers need forward-looking liquidity visibility, not only past transactions.	Forecast enterprise inflows/outflows with confidence bands and seasonal context.
Multi-source data	Financial records alone are thin or unavailable for many rural enterprises.	Fuse savings/loan records, UPI-like proxies, commodity trends, climate signals and sector patterns.
Early warning system	Risks must be flagged before repayment default, stock loss or business slowdown occurs.	Trigger LOW / MODERATE / HIGH risk alerts with root-cause explanation and intervention date.
Low-network + privacy constraints	Rural usage needs offline entry and privacy-conscious proxy signals.	Offline-first PWA/app, anonymized aggregate signals, consent-based data flow.
Actionable insights	A red flag alone is not useful for a field officer or entrepreneur.	Risk-to-Rescue engine suggests the smallest corrective action to stabilize cash flow.

Why the current approach fails

1 Fragmented data

Financial records, repayment behaviour, market prices, weather and UPI proxy signals remain in different silos.

2 Thin credit history

Many rural enterprises cannot prove repayment capacity because their cash flow is seasonal and informal.

3 Late intervention

Field officers usually see stress after delayed repayment, low stock turnover or liquidity shortage.

4 No “rescue” action

Most tools classify risk, but do not calculate what action can prevent the risk from becoming a default.

Core insight: the winning system must move from “risk identification” to “risk prevention”.

Innovation Pillars — What Makes It Stand Out

These features make the idea more than a generic credit-score or dashboard application.

Novelty pillar	What it does	Why it matters for NABARD
1. Enterprise Financial Digital Twin	Creates a simulated financial copy of each enterprise to test future cash-flow scenarios.	Enables forward planning rather than backward-looking credit evaluation.
2. Resilience Gap Metric	Calculates the minimum support needed to keep cash flow positive over 90/180 days.	Turns risk into a measurable intervention amount.
3. Risk-to-Rescue Engine	Compares multiple corrective actions and recommends the most practical option.	Field officers get “next best action”, not just a red flag.
4. Village / FPO Shock Radar	Detects correlated risk across many enterprises in a location or sector.	Supports group-level action such as bulk procurement or local advisory.
5. Trust & Confidence Passport	Shows data quality, missing fields, forecast confidence and explanation drivers.	Improves transparency and avoids blind trust in AI outputs.

“We do not only predict risk — we calculate how to prevent it.”

End-to-End Process Flow

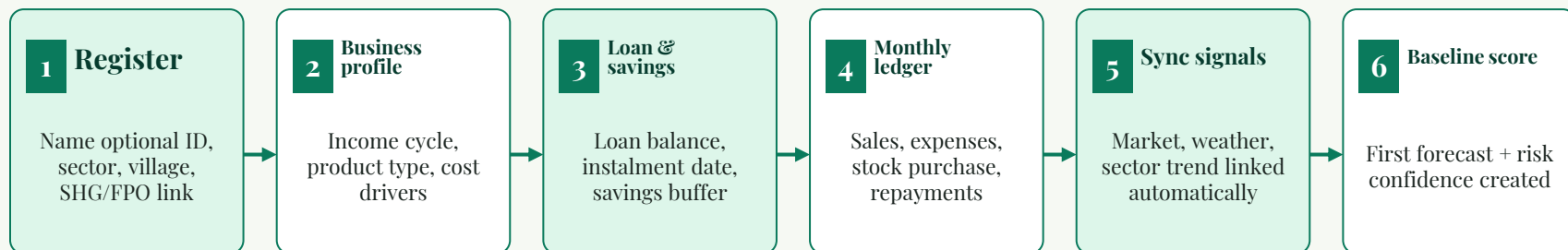
Sno	Process stage	Inputs	AI / Logic used	Output
1	Onboard enterprise	Sector, location, business type, loan/savings info	Data validation + profile template	Enterprise profile and baseline risk record
2	Capture transactions	Income, expenses, stock purchase, repayments, UPI proxy aggregates	Offline sync + anomaly checks	Clean monthly cash-flow ledger
3	Add external signals	Rainfall, temperature, commodity/raw-material prices, demand seasonality	Sector mapping + shock indexing	Market-climate risk features
4	Forecast cash flow	Ledger + proxy signals + seasonality	Quantile regression / XGBoost forecast	3–6 month forecast with confidence band
5	Flag risk	Forecast gap, repayment timing, shock index, balance trend	Risk classifier + SHAP explanation	Low/Moderate/High flag with top causes
6	Recommend rescue	Risk causes + available interventions	Scenario simulator + optimization rules	Minimum intervention plan and priority list
7	Track outcome	Officer action, repayment, enterprise response	Feedback learning loop	Improved future predictions and impact evidence

Data Layer and Privacy-by-Design

Data category	Examples captured	Purpose in model	Privacy handling
Financial records	Savings, loan balance, repayments, account balance if available	Baseline repayment capacity and liquidity trend	Consent-based; minimal fields only
Digital payment proxies	UPI inflow/outflow trend, transaction count, frequency, seasonality	Sales momentum and business activity pattern	Aggregated values; no payer/payee personal details
Market intelligence	Commodity price, input/raw material cost, local demand seasonality	Detect stress from price or demand shocks	Public/sector-level data
Climate signals	Rainfall deficit/excess, temperature anomaly, drought/flood risk	Predict income/expense disruptions by sector/location	Location-level only; no personal data
Field officer observations	Stock level, business continuity, repayment difficulty, event notes	Human verification and model correction	Role-based access with audit logs



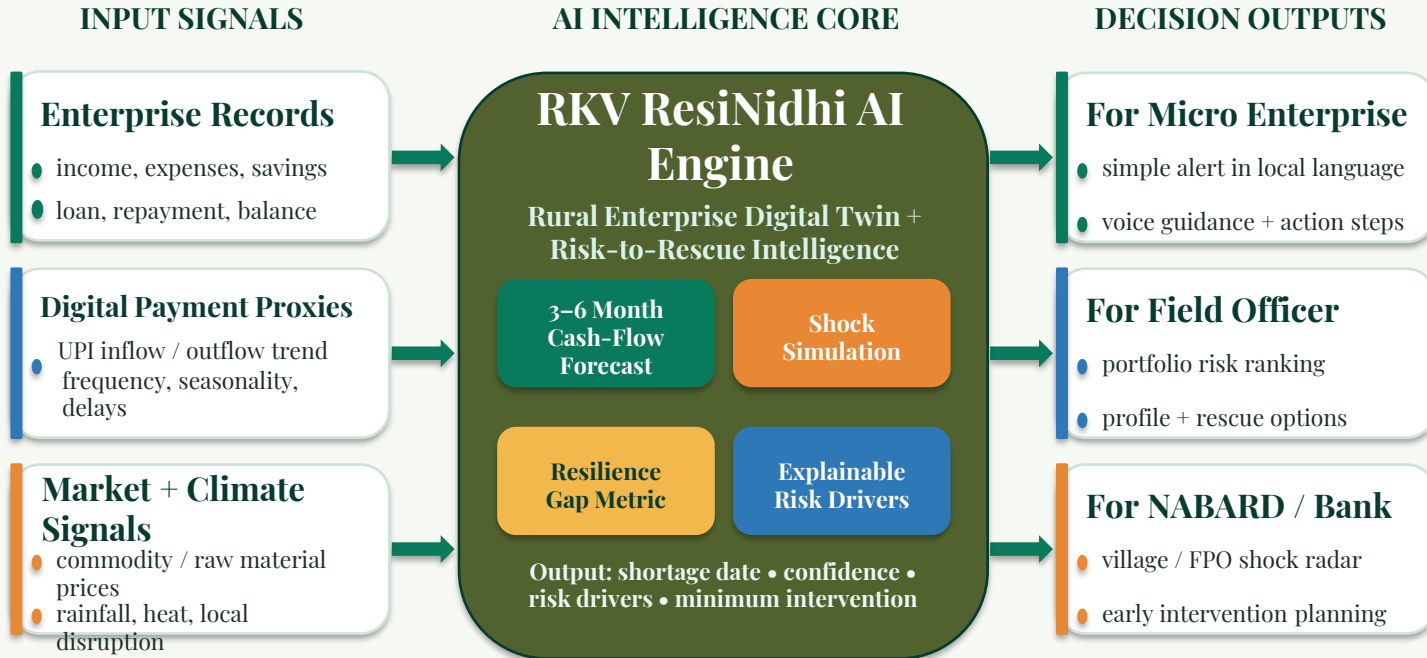
Enterprise Onboarding Flow



Offline-first design choice	Why it is required
Local device storage	Field officer can complete data entry even without network.
Background sync	Data uploads automatically when connectivity returns.
Voice / regional language entry	Supports entrepreneurs who are not comfortable with typed English forms.
Validation prompts	Reduces wrong entries such as negative expenses, missing repayment dates or inconsistent values.

Solution in One View

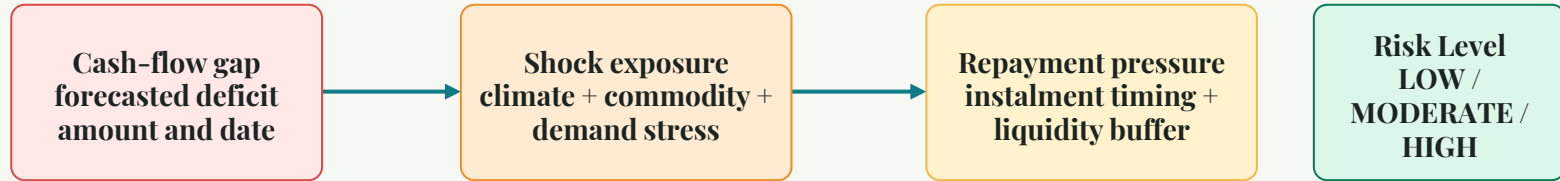
RKV ResiNidhi AI converts fragmented rural signals into cash-flow forecasts, risk explanations and practical rescue actions.



Cash-Flow Forecasting Engine

Module	Method	Reason for choosing it	Prototype output
Data cleaning	Rule-based validation + missing value treatment	Round 1/2 mock data may be imperfect and sparse.	Clean ledger and confidence level
Feature engineering	Seasonality, repayment calendar, shock index, moving averages	Captures rural business cycles better than raw amounts.	Model-ready feature table
Forecast model	XGBoost / LightGBM quantile regression	Works well with tabular, small and mixed proxy datasets.	Expected cash flow + low/high band
Cold-start model	Sector + district cohort benchmark	New enterprises may have limited historical data.	Baseline forecast using similar enterprises
Explainability	SHAP-style contribution ranking	Field officers need reason codes, not black-box output.	Top drivers: e.g., fodder price, low inflow, repayment date

Risk Scoring Logic – From Signal to Alert



Risk level	Trigger condition	Displayed explanation	Recommended priority
LOW	Stable forecast; expected balance remains positive	Seasonal dip is within normal historical range.	Monitor monthly
MODERATE	Short-term cash stress or repayment pressure visible	UPI inflow dropped and repayment date overlaps low-sales period.	Field call / advisory
HIGH	Projected negative cash position or compounded external shock	Market price fall + climate shock + weak liquidity buffer.	Immediate rescue action

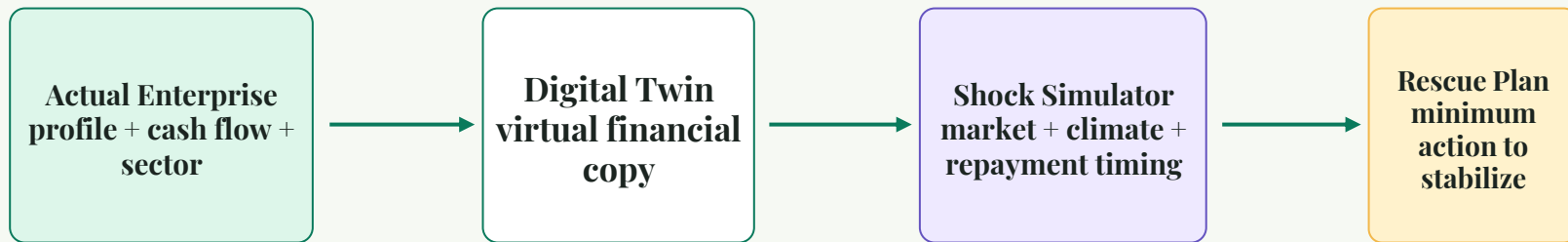


Every alert must show: risk level, forecast confidence, critical date, top three drivers and suggested next action.



Rural Enterprise Digital Twin

The digital twin simulates “what-if” shocks before they damage the enterprise.



Risk-to-Rescue Decision Table

The system does not approve loans automatically; it recommends explainable options for human review.

Possible rescue action	When recommended	Predicted result	Human control
Bridge working-capital support	Temporary cash shortage but business fundamentals are stable	Resilience Gap reduces; risk may shift High → Moderate/Low	Bank/field officer approves amount
EMI date adjustment	Cash inflow is delayed but expected soon	Prevents technical repayment stress during seasonal dip	Lender policy decides eligibility
Reduce/restock later	Inventory purchase creates cash pressure during low sales	Improves short-term liquidity without new credit	Enterprise accepts suggestion
Group/FPO procurement support	Many enterprises face same input-price shock	Bulk buying reduces cost pressure for cluster	FPO/NABARD partner coordinates
Field verification visit	Data quality low or model confidence weak	Prevents wrong decision from incomplete information	Field officer validates ground reality

Village / FPO Shock Radar

Detects cluster-level risk when many enterprises are affected by the same external factor.

Cluster signal	Example detection	Recommended group-level response
Dairy enterprises in same village	17/42 enterprises show rising fodder-cost stress	Bulk fodder procurement through FPO or local partner
Retailers near same market	UPI inflow proxy reduces for two consecutive weeks	Field officer checks market closure / demand disruption
Poultry units in same block	Feed price rise + local disease-risk note	Advisory + temporary working-capital review
Food processors in crop belt	Raw material price volatility after weather event	Inventory planning and price-risk guidance

HIGH CLUSTER RISK

Dairy • Village A
Common driver: fodder cost
Affected portfolio: ₹24.6L

MODERATE WATCH

Retail • Market B
Common driver: demand dip
Action: officer visit

LOW RISK

Handicrafts • Cluster C
Driver stable
Action: monthly monitoring



Micro Enterprise Experience

Screen / feature	What the entrepreneur sees	Why it helps
Daily / weekly entry	Income, expenses, stock purchase and loan payment entry in local language.	Creates a usable cash-flow history without complex accounting.
My Cash Flow	Next 3 months expected inflow, expenses and cash balance.	Helps business owner plan purchase and repayment.
Risk alert	Plain-language warning: “October may be difficult because expenses are rising.”	Avoids confusing technical risk terms.
Action suggestion	Delay restocking, save minimum buffer, discuss EMI date, request officer support.	Turns alert into a practical next step.
Trust Passport	Data quality, missing entries and confidence level.	Shows whether the alert is strong or needs verification.

Sample message: “Moderate risk for next month. Milk income is stable, but fodder expense is rising. Keep ₹5,000 buffer and avoid large stock purchase before 15 Oct.”

Offline entry • Voice support • Local language • Simple visual alerts

Field Officer Experience

Field officer module	Information shown	Decision supported
Portfolio risk list	All assigned enterprises sorted by High/Moderate/Low risk.	Prioritize urgent cases instead of manual follow-up.
Enterprise profile	Business sector, loan/savings status, recent cash-flow trend, data quality.	Understand context before intervention.
Risk explanation panel	Top three drivers such as input cost, repayment pressure, low inflow.	Avoid blind dependence on AI score.
Rescue recommendation	Bridge support, EMI shift, advisory, group procurement or verification visit.	Take practical corrective action.
Outcome tracker	Action taken, current status, repayment improvement, confidence update.	Measure whether intervention worked.

Priority	Enterprise	Risk	Reason	Next action
1	Lakshmi Dairy	HIGH	Fodder cost + EMI overlap	Bridge support / EMI discussion
2	Ravi Poultry	HIGH	Feed price shock	Cluster advisory
3	Meena Retail	MODERATE	Demand dip	Field call
4	Kavya Handicrafts	LOW	Stable orders	Monthly monitor

Feasibility, Risks and Mitigation

Risk / challenge	Impact if ignored	Mitigation in our design
Data sparsity	New enterprises may not have enough historical records.	Use sector/district cohort baseline and clearly show confidence level.
Incorrect manual entries	Wrong forecasts and unfair risk flags.	Validation rules, field verification and missing-data warnings.
Overdependence on AI	False confidence may harm borrower decisions.	Human-in-the-loop approval and explainable reason codes.
Privacy concerns	Users may not trust the platform.	Use aggregated proxies, consent-based sharing, role access and audit logs.
Connectivity limitations	Data entry and alerts may fail in rural areas.	Offline-first local storage and automatic sync.
Model bias across sectors	One model may unfairly judge different businesses.	Sector-specific features and separate thresholds for dairy, retail, poultry etc.



Conclusion — Rural Finance Intelligence

RKV ResiNidhi AI converts fragmented rural enterprise signals into forecasts, explanations and intervention decisions before financial stress becomes default risk.

FINAL TAKEAWAY

Not just “who is risky?” — RKV ResiNidhi AI answers: why risk will happen, when it will happen, and what minimum action can prevent it.

NABARD objective	How RaKVa Setu AI delivers it	Round 2 demo proof
3–6 month cash-flow prediction	Forecast future income, expenses and cash position using enterprise records + UPI proxies + market/climate signals.	Forecast dashboard with confidence range for dairy and rural retail.
Early risk detection	Classifies Low / Moderate / High risk using cash-gap timing, repayment pressure and external shock exposure.	Risk panel showing top three drivers and critical shortage date.
Actionable intervention	Calculates the Resilience Gap and compares rescue options such as bridge credit, repayment shift and group	Risk-to-Rescue table with before/after risk level.
Low-network rural usability	Offline-first app with simple data entry, voice/local-language support and sync when connectivity returns.	Mobile UI flow with offline data capture and field officer review.
Scalable public-good value	Village/FPO Shock Radar detects correlated risk across sectors so NABARD can act at portfolio level.	Cluster alert demo showing common shock and recommended group response.

Innovation

Digital Twin + Resilience Gap + Risk-to-Rescue

Feasibility

Prototype with mock datasets, two sectors and explainable AI

Impact

Earlier intervention, stronger credit appraisal and safer lending

Thank you!

